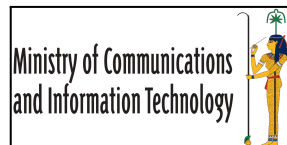




DIGITAL EGYPT PIONEERS INITIATIVE (DEPI)

Technical Catalog

Round 5
(Jul 2026-Feb 2027)





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Introduction

DEPI includes 7 technology tracks (specialties) that cover highly demanded jobs. Those tracks are:

1. Software Development
2. Digital Marketing & Art
3. AI & Data Science
4. Data Analytics
5. Infrastructure & Security
6. Management & ERP
7. Digital Game Creation Track

Each technology track offers training courses for technologies used by many companies in that specific track, and each course covers a potential job opportunity (job profile). The objectives of studying each job profile are

- Provide trainees with the necessary technical background and experience needed to excel in specific technologies that match trainee's capabilities and aspirations. This includes all the necessary prerequisites.
- Guide trainees on how to develop a professional and product quality module.
- Explain different aspects of the product development life cycle (planning, design, development, integration, and testing) that help trainees succeed in job market.
- Involve trainees in hands-on experience and training that deepen their understanding of the introduced technologies.

SOFTWARE DEVELOPMENT TRACK

The software track offers training courses for several web development technologies (frontend/backend) across different platforms, mobile development, cloud application development, DevOps, and software testing.

Software Development

DevOps Engineer
React Frontend Web Developer
Low-Code/No-Code Application Developer
Full Stack .Net Web Developer
Go Blockchain Developer
Cross Platform Mobile App Developer
Software Tester

SOFTWARE DEVELOPMENT TRACK

1. DevOps Engineer

A DevOps Engineer is responsible for smoothing the operation of a company's IT infrastructure. A DevOps Engineer's roles and responsibilities are a combination of technical and management roles. DevOps Engineers build and set up new development tools and infrastructure, implement integrations requested by customers, deploy updates and fixes, develop scripts to automate visualization, design procedures for system troubleshooting and maintenance, and optimize software development processes.

The technical study plan proposed for a DevOps Engineer is provided below:

Contents
Prompt Engineering
DevOps Essentials & Perquisites
Configuration Management using Ansible
Using Git for DevOps
Containerization basics using Docker
Docker Advanced
Continuous Integration using Jenkins
Introduction to Kubernetes
Continuous Monitoring using Prometheus
Cloud for DevOps
Terraform
Application Deployment (Resource Dependencies,....)
web server configurations
Capstone Project

SOFTWARE DEVELOPMENT TRACK

2. React Frontend Web Developer

This profile is equipped with comprehensive knowledge of web development essentials such as HTML, CSS, and JavaScript, and specialized in building React applications. With proficiency in Git and GitHub for version control, NodeJS for server-side scripting, Upgrades students from traditional Node.js/Express development to a modern, scalable full-stack architecture using NestJS for backend and Next.js for frontend. Also, an understanding of containerization with Docker and Kubernetes. The culmination of these skills is demonstrated in a capstone project, bringing React and NodeJS together to produce a real-world web application.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
HTML5 Essentials
CSS Essentials Principles of UX/UI Design
JavaScript
Typescript
Bootstrap
Getting Started with Git and GitHub
React
Code style, patterns and best practices
Introduction to Backend: NodeJS + Express
Introduction to NestJS & Structured Backend
Next.js Fundamentals & Frontend Integration
Containerization basics using Docker
Functional Documentation & Unit Testing
Capstone Project



SOFTWARE DEVELOPMENT TRACK

3. Low-Code/No-Code Application Developer

This profile enables learners to rapidly transform business needs into working digital solutions without relying heavily on traditional coding. It builds practical delivery skills, client communication, and portfolio-ready project experience. Learners graduate ready to support organizations with faster automation, internal tools, and business applications.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
No-Code Foundations, Client Discovery & UX Planning
No-Code Data Systems & Forms
No-Code Apps, Portals & Dashboards
No-Code Automation & Delivery Sprint
Low-Code Foundations & Visual App Building
Backend-Connected Apps & Authentication
Admin Panels, Internal Tools & Operations Apps
Integrations, Automation & Freelancer Delivery Sprint
Capstone Project

SOFTWARE DEVELOPMENT TRACK

4. Full Stack .Net Web Developer

Rooted in .NET technologies, this profile reflects a solid foundation for data-driven applications. With a strong grasp of C#, the candidate is equipped with robust back-end development. Front-end proficiency is ensured through HTML5, JavaScript, and CSS3, creating a full-stack capability. Experience with the .NET Core Web API and MVC frameworks enables the creation of scalable, modern web services and applications, completing a versatile skill set for comprehensive .NET development.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Introduction to SQL Server Programming
Getting Started with Git and GitHub
Programming in C#
Programming in HTML5, JavaScript & CSS3
Functional Documentation & Unit Testing
Containerization basics using Docker
.Net Core Web API
Developing .NET Core MVC Web Applications
Capstone Project

SOFTWARE DEVELOPMENT TRACK

5. Go Blockchain Developer

This profile equips learners with practical skills to build modern software systems using Go. Learners are prepared to become capable backend developers who can build reliable, efficient, and scalable software solutions. The profile strengthens problem-solving, system design thinking, and practical development confidence through hands-on implementation. Learners graduate with the ability to contribute to real backend systems and modern application development projects.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Web & Frontend Foundations
React Frontend Development
Go Programming Fundamentals
Intermediate Go & Concurrency
Backend Development with Go
Blockchain Fundamentals
Blockchain Development using Go
Capstone Project

SOFTWARE DEVELOPMENT TRACK

6. Cross Platform Mobile App Developer

A cross platform Mobile Application Developer is responsible for designing, developing, and maintaining a cross mobile application with highly detailed cross platform mobile development content. The courses in this profile cover a range of technologies that a Mobile Application Developer should have a strong understanding of and be able to use these technologies such as flutter technology to design and develop responsive and user-friendly mobile applications.

The technical study plan proposed for a cross platform mobile developer Associate is provided below:

Contents
Prompt Engineering
Dart Fundamentals For Flutter
Flutter Fundamentals
Flutter Advanced Topics
Firebase
UX/UI Basics
Getting Started with Git and GitHub
Functional Documentation & Unit Testing
Mobile Deployment for android and IOS
Capstone Project

SOFTWARE DEVELOPMENT TRACK

7. Software Tester

Software testing is a crucial part of the software development lifecycle, and a Software Tester is responsible for ensuring that software products meet quality standards. A successful Software Tester should have a strong understanding of the topics covered in the courses in this profile and be able to use them to design and execute effective testing strategies.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Java programming fundamentals
Fundamentals Of Testing (ISTQB 4)
Testing Throughout the Software Development Life Cycle (ISTQB 4)
Static Testing Techniques (Jira)
Test Tools (postman, selenium)
Test Analysis & Design
Managing Test Activities (Database and Agile)
Capstone Project

DIGITAL MARKETING & ARTS TRACK

Do you want to turn your love of art into a job? This program teaches you how! You'll learn how to create amazing things and how to sell them online.

Here's what you'll learn:

- Create stunning digital art and visuals: Learn AI in Digital Arts, motion graphics, and video editing to produce creative content for today's digital world.
- Design websites and apps: Learn UI/UX design to create experiences that are beautiful and easy to use.
- Promote your work online: Learn digital marketing strategies to help your projects reach the right audience and grow your personal brand.

We'll help you become a creative professional with the skills to design, create, and market digital content for today's fast-growing creative industry.

Ready to unleash your creativity and build a successful career? Join the Digital Marketing & Arts track!

**Digital
Marketing &
Arts**

- 1- Digital Marketing Specialist
- 2- UI/UX Web Designer
- 3- Motion Graphics Designer
- 4- Media Production for Influencers
- 5- AI in Digital Arts

DIGITAL MARKETING & ARTS TRACK

1. Digital Marketing Specialist

The Digital Marketing Specialist Profile equips learners with the skills to build and manage modern marketing campaigns that drive visibility, engagement, and measurable business growth. Through a hands-on approach, learners develop integrated marketing strategies combining content creation, social media, SEO, online advertising, and email marketing. They gain practical experience in audience targeting, campaign planning, content creation, and performance optimization using tools like Google Analytics, Google Ads, and Meta Ads Manager.

The program emphasizes data-driven marketing, enabling learners to analyze performance, measure ROI, and continuously improve campaigns. Through real-world projects and a capstone, participants build practical expertise in executing campaigns, managing platforms, and aligning marketing strategies with business goals.

Graduates are prepared to pursue careers as Digital Marketing Specialists, Social Media Managers, SEO Specialists, Content Marketing Specialists, Performance Marketing Executives, or Digital Campaign Managers, helping brands grow their online presence and achieve measurable results.

The technical study plan proposed for this job profile is provided below:

Contents
Foundations of Marketing & Digital Marketing
Content Marketing & Content Strategy Development
Social Media Marketing Platforms
Search Engine Optimization
Email Marketing
Google Ads
Mobile Marketing
Performance Measurement & Reporting
Affiliate Marketing
Allocating Budget

DIGITAL MARKETING & ARTS TRACK

2. UI & UX Web Designer

The UI/UX Design Professional Profile prepares learners to build modern digital products that are intuitive, visually engaging, and aligned with real user needs. In today's digital economy, businesses rely heavily on well-designed websites and mobile applications, making UI/UX design skills among the most in-demand capabilities in the technology and product development industries.

Throughout this hands-on UI/UX design course, participants learn how to research user behavior, define user journeys, and transform ideas into structured wireframes and interactive prototypes. Using industry tools such as Figma, Adobe XD, Adobe Photoshop, and Adobe Illustrator, learners develop the practical skills required to design responsive interfaces, create design systems, conduct usability testing, and collaborate effectively with developers during the product development process. The program also includes real-world design projects and a capstone experience that helps participants build a professional portfolio aligned with industry expectations.

Graduates can work as UI/UX Designers, Product Designers, Interaction Designers, UX Researchers, Web Designers, or Visual Interface Designers, contributing to the creation of digital platforms, mobile apps, and user-centered product experiences for companies, startups, and digital agencies.

The technical study plan proposed for this job profile is provided below:

Contents
UI-UX Fundamentals
Adobe Photoshop (for Web)
Adobe Illustrator (for Web)
Applying UI-UX using Adobe XD
Figma

DIGITAL MARKETING & ARTS TRACK

3. Motion Graphics Designer

The Motion Graphic Designer Profile prepares learners to create engaging animated visual content for digital media, marketing campaigns, and online platforms. As video becomes a dominant communication tool, this program equips participants with in-demand motion graphics and video animation skills for careers in media companies, agencies, startups, and digital brands.

Through a hands-on approach, learners explore animation fundamentals, visual storytelling, and motion design, covering the full production workflow from concept development and storyboarding to asset design and animation. Participants gain practical experience creating social media animations, explainer videos, promotional content, and branded motion graphics using tools such as Adobe After Effects and Adobe Premiere Pro, alongside Adobe Illustrator and Adobe Photoshop.

Graduates are prepared for roles such as Junior Motion Graphic Designer, Video Editor, Social Media Video Creator, Explainer Video Designer, YouTube Video Editor, or Animated Content Creator, contributing to the production of impactful visual content across digital platforms.

The technical study plan proposed for this job profile is provided below:

Contents
Adobe Illustrator (for Motion)
Adobe After Effects
Element 3D
Adobe Premiere Pro Fundamentals
Adobe Audition
Adobe Character Animator

Prerequisite:

Portfolio or Behance link for Photoshop & Illustrator Fundamentals

DIGITAL MARKETING & ARTS TRACK

4. Media Production Influencer

The Media Production Influencer Profile prepares learners to produce engaging digital media content for today's creator economy. As video and multimedia dominate social platforms, this program equips participants with in-demand skills to plan, produce, and edit high-quality visual and audio content for brands, businesses, and online creators.

Through a hands-on approach, learners cover the full content creation workflow, from concept development and visual storytelling to video editing, motion graphics, and audio production. Participants gain practical experience creating short-form videos, social media content, YouTube videos, and promotional media using tools such as Adobe Premiere Pro, Adobe After Effects, Adobe Photoshop, and Adobe Audition, while exploring audience engagement, digital influence strategies, and AI-supported content production with prompt engineering.

Graduates are prepared for roles such as Video Editor, Social Media Video Editor, Content Creator, UGC Content Creator, YouTube Video Editor, Reels and Shorts Video Editor, Podcast Video Editor, or Digital Content Producer, supporting brands, influencers, agencies, and digital platforms with impactful multimedia content.

The technical study plan proposed for this job profile is provided below:

Contents
Adobe Photoshop for Multimedia
Adobe Premiere Pro
Adobe After Effects
Adobe Audition
How to Influence

DIGITAL MARKETING & ARTS TRACK

5. AI in Digital Arts

The AI Digital Arts & Content Creation Profile prepares learners to create high-impact visual and multimedia content using cutting-edge AI tools. As artificial intelligence rapidly transforms the creative industry, professionals who can combine creativity with AI-driven production are among the fastest-growing and most in-demand talents across digital agencies, content studios, and online platforms.

Through hands-on training, learners master the full AI content creation workflow, including prompt engineering, image generation, video production, motion design, and audio integration. Participants gain practical experience producing social media content, branded visuals, and cinematic videos while applying modern AI trends in scalability, speed, and content personalization.

With project-based learning and a final portfolio, learners develop the ability to deliver complete AI-powered projects aligned with market needs and client expectations.

Graduates are prepared for roles such as AI Content Creator, AI Digital Artist, Video Creator, or Freelance Creative, supporting brands in producing scalable, engaging, and trend-driven digital content.

The technical study plan proposed for this job profile is provided below:

Contents
AI Creative Strategy & Prompt Engineering
AI Image Generation & Digital Art Production
AI Video Creation & Cinematic Content Development
Motion Graphics, Audio & Multimedia Integration
AI Content Production, Freelancing & Portfolio Development

AI & DATA SCIENCE TRACK

This track primarily focuses on applying data science and machine learning techniques to solve real-world challenges and problems and get familiar with a wide range of topics related to this field, for example: machine learning, deep learning, language processing, and others.

By the completion of the course, you will have

- The practical experience needed to use artificial intelligence to solve problems.
- The ability to deal with data in its various forms, collect it, extract data in appropriate forms that fit processing requirements.
- The ability to understand and analyze data, and extract important information from it, and thus provide recommendations to companies in various fields.
- Effectively implement and manage large-scale and complex AI projects.
- The ability to evaluate and analyze problems as well as propose solutions related to the development of artificial intelligence systems.

AI & DATA SCIENCE

Agentic and Generative AI System Developer

Microsoft Machine Learning Engineer

Data Scientist

Microsoft Data Engineer

AWS Machine Learning Engineer

AI Automation Workflow with N8N Profesional

AI & DATA SCIENCE TRACK

1. Agentic AI and Generative AI System Developer

Trainees will learn the required skills to use Agentic AI and generative AI in their professional lives. Additionally, they will explore the fundamental ideas behind Agentic AI and generative AI and gain knowledge about new AI models. Furthermore, trainees will be introduced to the moral implications of employing Agentic AI and generative AI.

The technical study plan proposed for this job profile is provided below:

Contents
Generative AI Overview
Transformer & LLM Deep Dive
LLM Architecture, Behavior & Debugging
LLM Usage & API Integration
Advanced Prompting
Hugging Face Models
Open LLMs & OpenCLAW
RAG & Memory Systems
Agentic AI Fundamentals
Multi-Agent Systems
AI Apps & UI Development
Deployment & Monitoring
Capstone Project

Prerequisite: Good Experience in Machine Learning and Deep Learning

AI & DATA SCIENCE TRACK

2. Microsoft Machine Learning Engineer

Trainees will learn how to use Microsoft technology to translate the raw data gathered from various data pipelines into data science models that can be applied and scaled as needed. In addition, they will also develop algorithms and build programs that enable machines, computers, and robots to process incoming data and identify patterns.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Math (Statistics and Linear Algebra Basics)
Python Programming
Python for Data Science, AI & Development
Preprocessing & Visualization
Machine Learning Techniques
Deep Learning
Natural Language Processing with Attention Models
Computer Vision and Transfer Learning
Azure AI Fundamentals
Azure AI Engineer Associate
MLOps Tools, MLflow, and Hugging Face
Capstone Project

Applicants will be required to sign up for a Microsoft Azure account in order to complete this profile. The sign-up process may require a valid Credit card or “.edu” email.

AI & DATA SCIENCE TRACK

3. Data Scientist

This program is designed by IBM. A data scientist can leverage a unique set of skills and tools to help businesses derive value from their data. A data scientist is capable of mining the data for drivers of past and future performance, opportunities for new revenue, methods to increase efficiency, ways to improve the customer experience, and a host of other critical business insights.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
What is Data Science?
Tools for Data Science
Data Science Methodology
Python for Data Science, AI & Development
Python Project for Data Science
Databases and SQL for Data Science with Python
Data Analysis with Python
Data Visualization with Python
Machine Learning with Python
MLOps Tools, MLflow and Hugging Face
Capstone Project

AI & DATA SCIENCE TRACK

4. Microsoft Data Engineer

This program prepares learners to build and manage modern data systems that support analytics, AI, and business intelligence solutions. Participants develop practical skills in programming, database management, data pipelines, big data processing, and cloud-based data engineering using Microsoft Azure technologies. The program focuses on scalable data architectures, deployment practices, and AI-driven data workflows through hands-on projects and real-world applications aligned with industry demands.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Intro to Data Engineering
Programming Essentials
SQL & Database Management
Advanced Python for Data Engineering
Data Pipelines
Big Data Processing
Microsoft Azure Data Engineer Associate
Deployment
AI for Data Engineers
Capstone project

AI & DATA SCIENCE TRACK

5. AWS Machine Learning Engineer

This program equips learners with the knowledge and practical skills required to design, build, and deploy modern machine learning and Generative AI solutions using cloud technologies. Through hands-on learning and real-world applications, participants develop expertise in data engineering, machine learning foundations, natural language processing, and AI solution development on Amazon Web Services. The program emphasizes industry-aligned practices, scalable AI workflows, and applied projects that prepare learners for professional AI and machine learning roles.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
AWS Academy Data Engineering
AWS Academy Machine Learning Foundations
AWS Academy Machine Learning for Natural Language Processing
AWS Academy Generative AI Foundations
Capstone Project

AI & DATA SCIENCE TRACK

6. AI Automation Workflow with N8N Professional

This program prepares learners to design intelligent digital workflows that combine AI capabilities with business automation to improve productivity, decision-making, and operational efficiency. The program emphasizes practical implementation, enabling participants to build scalable AI-driven solutions and intelligent agents for real-world organizational needs using n8n. Learners will gain hands-on experience in integrating AI into modern work environments while following best practices for security, governance, and responsible AI adoption.

The technical study plan proposed for this job profile is provided below:

Contents
AI Fundamentals & Generative AI
ML Literacy for Non-Engineers
Prompt Engineering
Advanced Prompting
AI Agents: Concepts to First Prototype
Automation Foundations with n8n
Advanced AI Automation with n8n
Agents at Scale (Slack, Web, Email)
Production, Security, Governance & Ethics
Applied Reference Workflows & Selling AI Internally
Capstone Projects

DATA ANALYTICS TRACK

Data analysis aims at getting useful insights using the given data to support the process of decision making. The field of data analysis has witnessed great importance in recent years due to the boom in technology and the availability of advanced tools and techniques that help provide better results and solutions to various organizations.

The educational objectives of the program are:

- Learning different ways of analyzing data.
- Applying statistical models and modern data science tools to understand and analyze data and solve business problems.
- Studying machine learning algorithms to make the right decisions and meet the challenges of modern business.
- Acquiring different skills, tools, and methodologies to explore past business performance and drive future planning.

Data
Analytics

Professional Data Analyst

Junior Data Analyst

DATA ANALYTICS TRACK

1. Professional Data Analyst

A professional data analyst is responsible for performing in-depth data analysis, integrating multiple data sources, and delivering actionable insights that support strategic and operational decisions. This intermediate-level training program builds on foundational knowledge by equipping participants with advanced skills in data preparation, analysis, and visualization using Python, SQL, Power BI, and Tableau. Learners develop the ability to handle complex datasets, apply advanced SQL queries, perform exploratory and statistical analysis in Python, and design interactive dashboards and data models that address real-world business problems. The program also covers data governance practices and prompt engineering to enhance analytical workflows. Through hands-on projects and a capstone experience, participants learn to formulate analytical questions, uncover deeper patterns and trends, and deliver well-structured, data-driven recommendations that improve business performance and decision-making processes.

The technical study plan proposed for this job profile is provided below:

Contents
Python Foundation
Data Preparation for Exploration using Python
Data cleaning process and organization of data using SQL
Data Analysis to Answer Questions
Advanced SQL
Data Visualization using Tableau
Introduction to Power BI
Extract Data from Different Sources in Power BI
Data Cleaning using Power Query
Introduction to DAX
Data Analysis and Visualization using Power BI



Dashboard and Report Development in Power BI
Data Modeling and Integration in Power BI
Publishing & Sharing Power BI Report
Data Refresh Mechanisms in Power BI
Data Governance
Prompt Engineering
Capstone Project

DATA ANALYTICS TRACK

2. Junior Data Analyst

A junior data analyst is responsible for supporting basic data analysis tasks by collecting, cleaning, and organizing data, and generating simple insights to aid decision-making. This beginner-level training program introduces participants to the fundamentals of data analytics, including data concepts, basic statistics, and data-driven thinking, along with hands-on skills in Excel, Power BI, and introductory SQL and Python. Learners are guided through essential processes such as data cleaning using Power Query, building simple visualizations, and creating basic dashboards and reports. The program also introduces foundational concepts in data governance and prompt engineering to build awareness of data quality and emerging AI tools. Through practical exercises and a capstone project, participants develop the ability to explore datasets, identify basic patterns and trends, and communicate insights effectively, preparing them for entry-level roles in data analysis and business intelligence.

The technical study plan proposed for this job profile is provided below:

Contents
Introduction to Data Analysis & Data Driven Decisions
Introduction to Power BI
Extract Data from Different Sources in Power BI
Data Cleaning using Power Query
Introduction to DAX
Data Analysis and Visualization using Power BI
Dashboard and Report Development in Power BI
Data Modeling and Integration in Power BI
Publishing & Sharing Power BI Report
Data Refresh Mechanisms in Power BI
Visualization using Tableau
Introduction to SQL
Introduction to Python



Data Governance
Prompt Engineering
Capstone Project

INFRASTRUCTURE & SECURITY TRACK

With the increase in the use of the Internet in the tasks of our daily lives (such as electronic banking services, online shopping, social media, etc.), personal data has become increasingly vulnerable to electronic attacks, as attackers use modern methods and technologies to overcome traditional security controls. This is why cyber security has become an important field and is in high demand.

The cyber security track includes the techniques and mechanisms for ensuring the security of computer systems, data, and networks and protecting them from various electronic attacks.

The educational objectives of the program are:

1. Developing the manpower required to protect Egypt from electronic threats, while ensuring that security priorities are aligned with the country's work needs and strategies
2. Empowering graduates with the technical knowledge and skills necessary to protect and defend computer systems and networks
3. Develop graduates who are able to identify, analyze, and remedy computer security breaches
4. Develop graduates who can plan, implement and monitor cyber security mechanisms to help ensure that IT assets are protected.

INFRASTRUCTURE AND SECURITY

1. Governance and Risk Compliance
2. Information Security Compliance & ISO 27001 Specialist
3. AWS Security Cloud Computing
4. Cyber Security Incident Response Analyst
5. Vulnerability Analyst and Penetration Tester

INFRASTRUCTURE & SECURITY TRACK

1. Governance and Risk Compliance

This course introduces the principles and practices of IT Governance, Risk Management, and Compliance, equipping learners with the knowledge to align technological operations with organizational objectives and regulatory requirements. Participants will explore frameworks, policies, and controls that strengthen resilience, ensure accountability, and support effective decision-making. Through practical examples and case studies, the course emphasizes how GRC integrates into daily operations to manage risks, safeguard information assets, and maintain compliance in a dynamic business environment.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Governance of Enterprise IT
IT Resources
IT Compliance
Risk Optimization

Prerequisite:

- Risk and Security Management knowledge.

INFRASTRUCTURE & SECURITY TRACK

2. Information Security Compliance & ISO 27001 Specialist

The PECB ISO/IEC 27001 Lead Implementer course is a comprehensive training program designed to equip participants with the practical knowledge and implementation skills required to establish, implement, maintain, and continually improve an Information Security Management System (ISMS) based on ISO/IEC 27001:2022.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Foundations of Information Security & ISO 27001 Fundamentals
ISO/IEC 27001 Requirements & Annex A Controls
ISMS Risk Management & Implementation
Performance Evaluation, Audit & Continual Improvement
ISO 27001:2022 Standard Implementation Workshop
Capstone Project

Prerequisite:

- Cybersecurity Basics
- Networking Essentials

INFRASTRUCTURE & SECURITY TRACK

3. AWS Security Cloud Computing

This Secure Cloud Computing course is an AWS-focused learning path that builds cloud fundamentals, architectural design skills, and practical security capabilities through four aligned modules: AWS Cloud Practitioner, Architecting on AWS, AWS Security Essentials, and Security Engineering on AWS. Learners progress from core AWS concepts and services to designing resilient architectures and implementing security controls, including identity and access management, monitoring and logging, incident response, and compliance practices. The program combines instructor-led explanation, real-world scenarios, and hands-on labs to help participants apply AWS best practices for designing and operating secure cloud environments.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Domain 1: AWS Cloud Practitioner
Domain 2: ARCH on AWS
Domain 3: Cloud Basics, Concepts, Design and Architecture
Domain 4: Secure Cloud Data & Secure Cloud Infrastructure
Domain 5: AWS Security Essentials
Domain 6: Security Engineering on AWS
Domain 7: Secure Cloud Applications & Operation
Domain 8: Cloud Standards, Laws and Regulations
Capstone Project

Prerequisite:

- Basic IT concepts
- Networking fundamentals
- Security fundamentals

INFRASTRUCTURE & SECURITY TRACK

4. Cyber Security Incident Response Analyst

A Cyber Security Incident Response Analyst is capable of the following; responding to incidents and threats tracked by the Incident Response Management System. Incidents may include phishing, network, email, IDS, or antivirus alerts, Performing log, vulnerability, or compliance analysis/remediation, Collection, collating, and secure handling of evidence for storage and/or processing, Drafting reports and security briefs as needed, and Participation in intra-departmental meetings and projects in support of Network, Infrastructure, and Support groups as assigned.

Through a structured progression of five core modules, participants will build expertise across network and operating system fundamentals, offensive and defensive security techniques, digital forensics, incident response frameworks, and SIEM-driven SOC operations. Each module concludes with a dedicated session on integrating artificial intelligence — including local LLMs via Ollama — into real-world SOC workflows.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Module 1: Network Fundamentals
Module 2: OS Fundamentals (Windows & Linux)
Module 3: Security & Attack Techniques
Module 4: Incident Response & Digital Forensics
Module 5: SIEM, SOC Operations & Automation
Capstone Project

Prerequisite:

- Network basics
- Familiarity with Windows and/or Linux operating systems
- General comfort with command-line interfaces

INFRASTRUCTURE & SECURITY TRACK

5. Vulnerability Analyst and Penetration Tester

A Vulnerability Analyst / Penetration Tester is capable of the following; Ensuring that all Network security tests are conducted within the framework set by Mastercard, Ensure that all test cases and tools are used appropriately for testing various network segment types, Ensuring that all help and guidance is provided to development teams on the vulnerabilities identified in their applications while performing security testing, Identify improvement areas in the security testing domain and implement learnings for the Globally located team, Coordinate with application development teams on their demands for security testing and provide a seamless experience with testing and reporting of penetration tests, and Working with a global team spread across time-zones and geographies.

The technical study plan proposed for this job profile is provided below:

Contents
Prompt Engineering
Fundamentals of Networking & prerequisites
Network Penetration Testing
Web Application Penetration Testing
Android Penetration Testing
Capstone Project

Prerequisite:

- Network basics

Management & ERP TRACK

The field of Enterprise Resource planning is concerned with the study of various techniques for organizing and managing the personnel and resources of companies with high efficiency, improving operations, achieving automation, and making decisions based on data and various scientific and technological methods that have boomed in recent years. The importance of studying this specialization lies in increasing the income of individuals and companies through various job possibilities opening up in the business and technology sectors, as institutions increasingly apply ERP systems locally and globally. Also, the experiences gained from the study contribute to the success of the institutions, and this success is represented by increasing their competitiveness with international companies.

Management & ERP

SAP S/4HANA MM Consultant
SAP S/4HANA FI Consultant
Odoo Application Consultant
Google Project Management

The educational objectives of the program are:

- The study provides a comprehensive understanding of administrative processes and scientific methods for improving them, and enhances problem-solving skills using various algorithms.
- The study equips individuals with the necessary skills to adapt to digital transformation and collaborate and communicate effectively within multidisciplinary teams.
- Enabling organizations to manage their daily tasks and internal and external resources with high efficiency.



- Learn about and use the different technologies used globally, such as MS Dynamics 365, and Odoo.

Management & ERP TRACK

1. SAP S/4HANA MM Consultant

SAP S/4HANA Materials Management (MM) Consultant. It covers the end-to-end procurement and inventory management processes including purchasing, goods receipt, invoice verification, material master data, vendor management, and integration with Finance and other modules.

participants will be able to support SAP MM implementation, configuration, and day-to-day business operations in S/4HANA.

The technical study plan proposed for this job profile is provided below:

Contents
Introduction to SAP S/4HANA ERP
Business user role in MM and purchasing
Enterprise Structure Configuration
Process Configuration
Master Data Configuration
Process Execution
Prompt Engineering
Capstone Project

Prerequisite: Graduate and year 4
Preassessment exam.

Management & ERP TRACK

2. SAP S/4HANA FI Consultant

SAP S/4HANA Finance (FI) Consultant with strong foundational knowledge in SAP FI configuration and business processes. Skilled in understanding enterprise structure, financial accounting integration.

Knowledgeable in configuring master data, General Ledger (GL), Accounts Payable (AP), Accounts Receivable (AR), Bank Accounting, and Asset Accounting within S/4HANA.

Contents
Introduction to SAP S/4HANA ERP
Business user role in GL, AR, AP, and Fixed Assets.
Enterprise Structure Configuration
Process Configuration
Master Data Configuration
Process Execution
Controlling (CO) Overview
Prompt Engineering
Capstone Project

Prerequisite: Graduate and year 4

Preassistentment exam

Management & ERP TRACK

3. Odoo Application Consultant

The target job for an Odoo Application Consultant involves designing, implementing, and deploying software for applications in the Odoo ERP suite. They specialize in developing strategies to optimize a company's management processes using Odoo tools and modules to ensure efficiency and smooth workflow. The primary qualifications to become an Odoo consultant include a bachelor's degree in computer science or a relevant field and experience working with the Odoo ERP suite. An Odoo consultant may collect and analyze business requirements, write business requirement documents, and other functional and technical documents. The technical study plan proposed for this job profile is provided below:

Contents
ERP Introduction and ERP Accounting Configuration
Finance module GL, AR, AP and banks
Advanced configuration Taxes, Currencies and Consolidation Concept
Supply Chain Purchasing and purchase order management
Supply chain inventory, warehouse and inventory operations
Supply chain manufacturing and production planning
Sales CRM & Sales Pipeline Management
Sales templates, invoicing and point of sales
Human resources Employees Management and Attendance Management
Human resources Employment Contracts & Organizational Chart
Artificial Intelligence (AI) Applications in Data Automation
Prompt Engineering
Capstone Project

Prerequisite: Graduate and year 4
Preassistent exam.

Management & ERP TRACK

4. Google Project Management

Google project management profile prepares the applicant to be a successful project manager whose roles and responsibilities include but not limited to :

- Being actively involved in the preparation, conceptualization, realization, and deploying of customer implementation projects.
- Enhancing the project management tools for end-users based on requirements.
- Attending meetings with the client and gathering information on their current business processes.
- Conducting business process analyses, needs assessments, and leading industry/corporate process implementation across all related applications5.
- Assisting developers in project planning and delivery.
- Assisting customers in performing integration and acceptance testing.

The technical study plan proposed for this job profile is provided below:

Contents
Foundations of Project Management
Project Initiation: Starting a Successful Project
Project Planning: Putting It All Together
Agile Project Management
Artificial Intelligence (AI)
Prompt Engineering
Capstone Project

Prerequisite: Graduate and year 4
Preassistentment exam

Digital Game Creation TRACK

This track is designed to upskill applicants in **game design and game development**, through giving them a structured, production-oriented training pathway into the professional game industry ecosystem. The program balances **theoretical game design frameworks** with **hands-on development using industry-standard tools** (such as Unity), and includes principles of **game production, prototyping, and play-testing**. Participants will finish the program with a **portfolio of playable prototypes** and a clear understanding of the roles and workflows in a modern game studio.

Each Profile focuses on building foundational production skills, portfolio-ready projects, and a practical understanding of industry workflows used in indie studios, beginner production environments, and interactive media projects.

Students will learn through practical exercises, guided projects, and collaborative production workflows that simulate real game development environments.

Digital Game
Creation

Immersive Game Designer
Game Developer

Digital Game Creation TRACK

1. Immersive Game Designer

The **Immersive Game Designer** introduces students to the visual production pipeline used in modern games and interactive experiences. Students learn how to create optimized 3D assets and interactive environments using professional tools such as **Blender** and **Unity**.

The Profile focuses on practical real-time workflows used in mobile games and introductory VR experiences. Students will understand how game assets move through the production pipeline, starting from concept planning and modeling, all the way to scene building, lighting, optimization, and final presentation inside Unity.

This Profile is designed to build strong foundational skills for students who are interested in 3D art, environment creation, interactive media, and beginner-level game production workflows.

The technical study plan proposed for this job profile is provided below:

Contents
Introduction to the game industry and production pipeline
3D modeling using Blender
Asset optimization for games
Importing and organizing assets in Unity
Mobile optimization awareness
Introduction to VR-ready environments
Real-time rendering workflows
collaboration between artists and developers
Capstone Game Design Project

Digital Game Creation TRACK

2. Game Developer

The Game Development profile equips learners with the essential knowledge and practical skills required to design, develop, optimize, and test modern mobile and 3D games. The program begins with an introduction to the game industry, production pipelines, and core game concepts, followed by programming fundamentals tailored for game creation. Students gain a strong foundation in mobile development and Android technologies, enabling them to build interactive and performance-efficient games. The profile also covers game loop architecture, rendering systems, 2D and 3D game development, gameplay programming, and mobile performance optimization techniques. In addition, learners explore Android integration, game monetization strategies, and professional testing and debugging practices to ensure high-quality gaming experiences across mobile platforms.

The technical study plan proposed for this job profile is provided below:

Content	
1	Game Industry & Production Pipeline
2	Core Game Concepts
3	Programming Basics for Games
4	Mobile Development Essentials
5	Introduction to Game Testing
6	Android Foundations for Game Development
7	Game Loop & Rendering Architecture
8	2D Game Development
9	Introduction to 3D Game Development



10	3D Gameplay Programming
11	Mobile Performance Optimization
12	Android Integration & Monetization
13	Testing & Debugging Mobile Games
14	Capstone Game Development Project

SKILLS TRACK

1. Skills

1.1. Personal skills

The soft skills track focuses on equipping learners with professional competencies that strengthen workplace effectiveness and client interactions. Participants develop career planning strategies, personal branding, communication, emotional intelligence, presentation skills, CV writing, and interview techniques.

#	Contents
1	Career Planning
2	Personal Branding & Networking
3	Communication & Emotional Intelligence
4	CV Writing & Interview Skills
5	Presentation Skills
6	Employment Coaching

SKILLS TRACK

1.2. Freelancing Skills & Coaching for Gigs

The Freelancing Module equips participants with the knowledge and practical skills needed to enter and succeed in the freelance market. It covers market awareness, identifying in-demand skills, and analyzing successful freelancers, followed by building a strong portfolio and defining a unique value proposition. Participants learn how to navigate freelancing platforms, create professional accounts, attract clients, and manage communication. The module also develops essential business skills such as pricing, budgeting, and writing effective proposals, resulting in a final project where participants present their portfolio, platform presence, and client-ready materials.

#	Contents
1	Planning for Freelancers
2	Portfolio Building
3	Freelancing Platforms
4	Budgeting, Pricing & Proposals
5	Negotiation Skills
6	Kickstart Your Freelance Profile
7	Coaching (working on platforms and achieving gigs)

SKILLS TRACK

1.3. Entrepreneurship Track

The Entrepreneurship Module equips participants with the mindset, tools, and skills to turn ideation into viable ventures. It covers entrepreneurship fundamentals, innovation, design thinking for problem-solving, solution validation with the Business Model Canvas, startup essentials like marketing, finance, and team management, and culminates in a structured project pitch with feedback. The focus is on practical, collaborative application.

#	Contents
1	Introduction to Entrepreneurship
2	Design Thinking Methodology
3	Business Model Canvas
4	Practical Application
5	Marketing for Startups
6	Financial Planning for Startups
7	Team Management
8	Pitching Skills
9	Final Project Presentation

SKILLS TRACK

2. English

The program develops practical English communication and professional skills for aspiring freelancers. Freelance Fundamentals builds core English client engagement abilities, including emails, pitches, interviews, negotiations, meetings, and feedback, with hands-on practice and portfolio-ready outputs. Client Engagement and Professional Interaction advances these English skills, covering presentations, contracts, proposals, expectation management, and Entrepreneurship. Graduates gain a practical English toolkit and portfolio assets to succeed in the global freelance market.

#	Contents
1	Freelancing Fundamentals
2	Client handling & Engagement
3	Platform presence and proposal strategy
4	Digital tasks and technical fluency
5	Strategic positioning and showcase delivery